Lindblad Master Equation: Generator of Markovian CPTP Maps

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Overview

Firstly, we introduce and motivate the most general properties of a quantum channel, namely it is linear and CPTP. We discuss notions of Markovian channels and its relation to divisibility. Secondly, we set up an open quantum system and the basic notations while also introducing the representations of a CPTP map by Stinespring's dilation and Kraus operators. Definitions to quantum dynamical semigroup is also given.
Next, we give the derivation to the GKSL equation, a generator of Markovian CPTP maps. Extra attention is given of the approximations used, determining in which physical systems do the GKSL equation hold. After that, properties to the GKSL equation are also discussed.
Lastly, a summary to all the above topics is given.

Quantum Channel

1. A quantum channel is *convex linear*, *completely positive* (*CP*), and *trace-preserving* (*TP*). Note that positivity is not enough, e.g. transpose map.

$$\boxed{\Phi(\lambda\rho_1 + (1-\lambda)\rho_2) = \lambda\Phi(\rho_1) + (1-\lambda)\Phi(\rho_2)} \quad \boxed{\Phi \otimes \mathbb{1}_d \succeq 0} \quad \boxed{\mathrm{Tr}\{\Phi(\rho)\} = \mathrm{Tr}\{\rho\}}$$

2. Adjoint map $\Phi^\dagger$: $\mathrm{Tr}\{\Phi(\rho)\cdot\sigma\} = \mathrm{Tr}\{\rho\cdot\Phi^\dagger(\sigma)\}$
3. A map Φ is CP iff $\Phi^\dagger$ is CP; Φ is TP iff $\Phi^\dagger$ is unital, i.e. $\Phi^\dagger(\mathbb{1}) = \mathbb{1}$.
4. Markovian channels are of the form $\mathrm{e}^{\mathcal{L}}$, where $\mathcal{L}$ is time-independent.
5. For time-dependent $\mathcal{L}$, the time-ordering operator is required. The channel formed by its exponential is called *infinitesimal divisible* and *time-dependent Markovian*.

Open Quantum System

Total system by including environment E evolves unitarily, use partial trace to obtain system evolution.
Representations of the channel include:

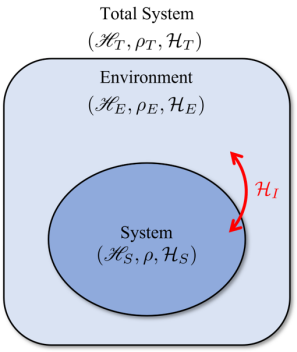
1. Stinespring's dilation: $\Phi(\rho) = \mathrm{Tr}_E\{U(\rho\otimes\rho_E)U^\dagger\}$, U is an isometry.
2. Quantum dynamical semigroup:

$$\begin{tikzcd}
\rho_T(t_0) = \rho(t_0)\otimes\rho_E \arrow[r, "\text{unitary evolution}"] \arrow[d, "\mathrm{Tr}_E"'] & \rho_T(t) = U_{t,t_0}\cdot\rho(t_0)\cdot U^\dagger_{t,t_0} \arrow[d, "\mathrm{Tr}_E"] \\
\rho(t_0) \arrow[r, "\text{dynamical map}"] & \rho(t) = V_{t,t_0}\cdot\rho(t_0)
\end{tikzcd}$$

- Dynamical semigroup: maps state to state.
- Quantum dynamical semigroup: a CPTP dynamical semigroup.

3. Kraus operator summation representation:

$$\Phi(\rho) = \sum_\alpha K_\alpha\rho K_\alpha^\dagger, \quad \sum_\alpha K_\alpha^\dagger K_\alpha = \mathbb{1}$$



Derivation

Short-Time Expansion

Stinespring's Dilation
$\rho(t) = \mathrm{Tr}_E\left\{U\rho_T(0)U^\dagger\right\}$
*Assume separable initial state
$\rho_T(0) = \rho(0)\otimes\rho_E$

Kraus Operators Summation Representation
$$\rho(t) = \sum_{\alpha=1}^{r} K_\alpha\rho(0)K_\alpha^\dagger$$

Compare with Taylor expansion. Find Ansatz for Kraus operators.
$$K_0 = \mathbb{1} + L_0\,\mathrm{d}t$$
$$K_\alpha = \sqrt{\mathrm{d}t}\sqrt{\gamma'_\alpha}\,L_\alpha$$
$$L_0 = -\frac{1}{2}\sum_{\alpha\geq 1}\gamma_\alpha L_\alpha^\dagger L_\alpha - i\mathcal{H}$$

Assume Markovian
$$\dot\rho(0) = \mathcal{L}\rho(0) \to \rho(t) = \mathrm{e}^{t\mathcal{L}}\rho(0)$$

Lindblad Form
$$\dot\rho(t) = -i[\mathcal{H},\rho(t)] + \sum_{\alpha\geq 1}\gamma_\alpha\left(L_\alpha\rho(t)L_\alpha^\dagger - \frac{1}{2}\left\{L_\alpha^\dagger L_\alpha, \rho(t)\right\}\right)$$

Change of Coordinates

GKS Form
$$\dot\rho(t) = -i[\mathcal{H},\rho(t)] + \sum_{i,j} G_{ij}\left(F_i\rho(t)F_j^\dagger - \frac{1}{2}\left\{F_j^\dagger F_i, \rho(t)\right\}\right)$$

Microscopic Derivation

Approximations used:
- Weak coupling;
- Markov approximation;
- Rotating wave approximation.

Implying that the GKSL equation is only valid when applying on systems with (1) weak system-environment interactions, (2) fast decay of excitation in environment.

Properties

1. Heisenberg picture: $\mathcal{L}^\dagger A = i\,[\mathcal{H}, A] + \sum_\alpha\gamma_\alpha\left(L_\alpha^\dagger A L_\alpha - \frac{1}{2}\left\{L_\alpha^\dagger L_\alpha, A\right\}\right)$
2. Since Lindblad form is non-unique, one can always choose traceless Lindblad operators L_α.
3. Purity reducing: for Hermitian Lindblad operators, we have $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Tr}\{\rho^2\} \leq 0$

References

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Microscopic Derivation to GKSL Equation

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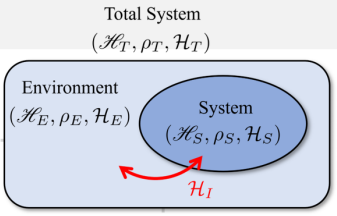
- System Setup (See picture on the right)



Total Hamiltonian: $\mathcal{H}_T = \mathcal{H}_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \mathcal{H}_E + \mathcal{H}_I$
Interaction Hamiltonian: $\mathcal{H}_I = \sum_i S_i \otimes E_i,\ S_i \in \mathcal{B}(\mathscr{H}_S),\ E_i \in \mathcal{B}(\mathscr{H}_E)$

Schrödinger's Picture

$$\dot{\rho}(t) = -i[\mathcal{H}(t), \rho(t)]$$
$$\dot{A}(t) = 0$$
$$\langle A \rangle = \mathrm{Tr}\left\{A(t)\rho(t)\right\}$$

Interaction Picture (denoted by hats)

$$\begin{cases} \hat{\rho}(t) = \mathrm{Ad}_{\exp\{it(\mathcal{H}_S\otimes\mathbb{1}_E + \mathbb{1}_S\otimes\mathcal{H}_E)\}}\rho(t) \\ \hat{A}(t) = \mathrm{Ad}_{\exp\{it(\mathcal{H}_S\otimes\mathbb{1}_E + \mathbb{1}_S\otimes\mathcal{H}_E)\}}A_0 \end{cases} \qquad \begin{cases} \frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}(t) = -i\left[\hat{\mathcal{H}}_I, \hat{\rho}(t)\right] \\ \frac{\mathrm{d}}{\mathrm{d}t}\hat{A}(t) = i\left[\mathcal{H}_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \mathcal{H}_E, \hat{A}(t)\right] \end{cases}$$

$$\langle A \rangle = \mathrm{Tr}\left\{A_0\rho(t)\right\} = \mathrm{Tr}\left\{\hat{A}(t)\hat{\rho}(t)\right\}$$

- Notation

Main derivation steps. Change of variables / Sub-steps / Extra information. Approximations.

- Derivation

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_T(t) = -i\left[\hat{\mathcal{H}}_I(t), \hat{\rho}_T(t)\right] \quad \text{(Liouville-von Neumann equation in interaction picture)}$$

$$\hat{\rho}_T(t) = \hat{\rho}_T(0) - i\int_0^t \left[\hat{\mathcal{H}}_I(s), \hat{\rho}_T(s)\right] \mathrm{d}s$$

weak coupling approximation:

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_T(t) = -i[\hat{\mathcal{H}}_I(t), \hat{\rho}_T(0)] - \int_0^t \left[\hat{\mathcal{H}}_T(t), \left[\hat{\mathcal{H}}_I(s), \hat{\rho}_T(t)\right]\right] \mathrm{d}s + \underbrace{\cancel{o(\hat{\mathcal{H}}_I^2)}}_{\approx 0}$$

$$\because \hat{\rho}_T(0) = \rho_T(0) = \rho(0) \otimes \rho_E$$
$$\therefore \left[\hat{\mathcal{H}}_I, \hat{\rho}_T(0)\right] = \sum_i \mathrm{Tr}_E\left\{\hat{E}_i\hat{\rho}_E\right\}\left[\hat{S}_i, \rho(0)\right]$$
$$\because \hat{\mathcal{H}}_T = \left(\hat{\mathcal{H}}_S + \sum_i \langle\hat{E}_i\rangle\hat{S}_i\right) \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \hat{\mathcal{H}}_E + \sum_i \hat{S}_i \otimes \left(\hat{E}_i - \langle\hat{E}_i\mathbb{1}_E\rangle\right)$$
$$\therefore \mathrm{Tr}_E\left\{\hat{E}_i\hat{\rho}_E\right\} = 0$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}(t) = -i\underbrace{\mathrm{Tr}\left\{[\hat{\mathcal{H}}_I(t), \hat{\rho}_T(0)]\right\}}_{=0} - \int_0^t \mathrm{Tr}_E\left\{\left[\hat{\mathcal{H}}_T(t), \left[\hat{\mathcal{H}}_I(s), \hat{\rho}_T(t)\right]\right]\right\} \mathrm{d}s$$

$s \mapsto t - s$

$$= -\int_0^t \mathrm{Tr}_E\left\{\left[\hat{\mathcal{H}}_T(t), \left[\hat{\mathcal{H}}_I(t-s), \hat{\rho}(t) \otimes \hat{\rho}_E\right]\right]\right\} \mathrm{d}s$$

weak coupling: $\hat{\rho}_T(t) = \hat{\rho}(t) \otimes \hat{\rho}_E$

$$= -\int_0^\infty \mathrm{Tr}_E\left\{\left[\hat{\mathcal{H}}_T(t), \left[\hat{\mathcal{H}}_I(t-s), \hat{\rho}(t) \otimes \hat{\rho}_E\right]\right]\right\} \mathrm{d}s$$

$\hat{\mathcal{H}}_I(t-s) \to 0$ as $s \gg \tau_E$ (Markov approximation: quick decay for excitation in E.)

$$= \int_0^\infty \mathrm{Tr}_E\left\{\underbrace{\hat{\mathcal{H}}_I(t-s)(\hat{\rho}(T) \otimes \hat{\rho}_E)\hat{\mathcal{H}}_I(t) - \hat{\mathcal{H}}_I(t)\hat{\mathcal{H}}_I(t-s)(\hat{\rho}(T) \otimes \hat{\rho}_E)}_{\text{(term)}} - \underbrace{(\hat{\rho}(T) \otimes \hat{\rho}_E)\hat{\mathcal{H}}_I(t-s)\hat{\mathcal{H}}_I(t) + \hat{\mathcal{H}}_I(t)(\hat{\rho}(T) \otimes \hat{\rho}_E)\hat{\mathcal{H}}_I(t-s)}_{\text{(term)}^\dagger}\right\} \mathrm{d}s$$

Let $\mathcal{H}_S = \sum_\varepsilon \varepsilon\Pi(\varepsilon)$ (spectral decomposition),

then define $S_i(\omega) := \sum_{\varepsilon'-\varepsilon=\omega} \Pi(\varepsilon)S_i\Pi(\varepsilon')$, with the properties: $\sum_\omega S_i(\omega) = S_i,\ S_i^\dagger(\omega) = S_i(\omega)$.

$$\hat{S}_i(\omega) = \mathrm{e}^{it\mathcal{H}_S}S_i(\omega)\mathrm{e}^{-it\mathcal{H}_S} = \sum_{n=0}^\infty \frac{(\mathrm{ad}_{i\mathcal{H}_S})^n S_i(\omega)}{n!}t^n = \sum_{n=0}^\infty \frac{(-i\omega t)^n S_i(\omega)}{n!} = \mathrm{e}^{-i\omega t}S_i(\omega)$$
$$\Rightarrow \hat{\mathcal{H}}_I = \sum_{i,\omega} \hat{S}_i(\omega) \otimes \hat{E}_i(t) = \sum_{i,\omega} \hat{S}_i^\dagger(\omega) \otimes \hat{E}_i(t) \quad (\hat{E}_i(t) = \mathrm{e}^{it\mathcal{H}_E}E_i\mathrm{e}^{-it\mathcal{H}_E})$$
$$= \sum_{i,\omega} \mathrm{e}^{-i\omega t}S_i(\omega) \otimes \hat{E}_i(t) = \sum_{i,\omega} \mathrm{e}^{i\omega t}S_i^\dagger(\omega) \otimes \hat{E}_i(t)$$

$$= \sum_{\omega,\omega'}\sum_{i,j} \mathrm{e}^{-(\omega'-\omega)t}\underbrace{\left(\int_0^\infty \mathrm{e}^{i\omega s}\mathrm{Tr}_E\left\{\hat{E}_j(t)\hat{E}_i(t-s)\hat{\rho}_E\right\} \mathrm{d}s\right)}_{= \int_0^\infty \mathrm{e}^{i\omega s}\mathrm{Tr}_E\left\{\hat{E}_j(s)\hat{E}_i(0)\hat{\rho}_E\right\} \mathrm{d}s =: \Gamma_{ij}(\omega)} \cdot \left[S_i(\omega)\hat{\rho}(t), \hat{S}_j^\dagger(\omega')\right] + \text{(Hermitian conjuagte)}$$

(By assuming the environment state is stationary, $[\mathcal{H}_E, \hat{\rho}_E] = 0$.)

$$= \sum_\omega \sum_{i,j}\left(\Gamma_{ij}(\omega)\left[S_i(\omega)\hat{\rho}(t), \hat{S}_j^\dagger(\omega)\right] + \Gamma_{ji}^\dagger(\omega)\left[S_i(\omega), \hat{\rho}(t)\hat{S}_j^\dagger(\omega)\right]\right)$$

rotating wave approximation: During τ_S where $\hat{\rho}(t)$ varies quickly, terms where $\omega' \neq \omega$ oscillates quickly, and they may be neglected.

$\Gamma_{ij}(\omega) = \frac{1}{2}\gamma_{ij}(\omega) + i\pi_{ij}(\omega),\ \gamma_{ij}(\omega) := \Gamma_{ij}(\omega) + \Gamma_{ji}^*(\omega) \succeq 0,\ i\pi_{ij}(\omega) := \Gamma_{ij}(\omega) - \Gamma_{ji}^*(\omega)$

$$= -i\left[\sum_\omega \sum_{i,j} \pi_{ij}(\omega)S_j^\dagger(\omega)S_i(\omega), \hat{\rho}(t)\right] + \sum_\omega \sum_{ij} \gamma_{ij}(\omega)\left(S_i(\omega)\hat{\rho}(t)S_j^\dagger(\omega) - \frac{1}{2}\left\{S_j^\dagger(\omega)S_i(\omega), \hat{\rho}(t)\right\}\right)$$

$$= -i\left[\hat{\mathcal{H}}_{LS}, \hat{\rho}(t)\right] + \mathcal{D}(\hat{\rho}(t))$$

Back to Schrödinger's picture, with $\hat{\mathcal{H}}_{LS}$ the *Lamb shift* Hamiltonian satisfying $[\mathcal{H}_S, \hat{\mathcal{H}}_{LS}] = 0$.

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho(t) = -i\left[\mathcal{H}_S + \hat{\mathcal{H}}_{LS}, \rho(t)\right] + \mathcal{D}(\rho(t)) \quad \blacksquare$$

- Results

Many approximations are used in the derivation above, they determine the preconditions on which physical systems must satisfy so that they evolve in time according to the GKSL equation. The approximations used and their implications are:

- **Weak coupling (Born approximation):**
  The interaction between the system and the environment is small, hence we can expand to $o(\hat{\mathcal{H}}_I^2)$. Moreover, the environment is only weakly affected by the interaction, hence $\hat{\rho}_T(t) = \hat{\rho}(t) \otimes \hat{\rho}_E$. Furthermore, we can even assume the environment to be stationary.
- **Markov approximation:**
  The excitation time scale of environment $\tau_E \ll \tau_S$, the time scale over which the system varies considerably. In other words, the *correlation function* $\mathrm{Tr}_E\left\{\hat{E}_j(s)\hat{E}_i(0)\hat{\rho}_E\right\}$ decays sufficiently fast compared to τ_S.
- **Rotating wave approximation.**